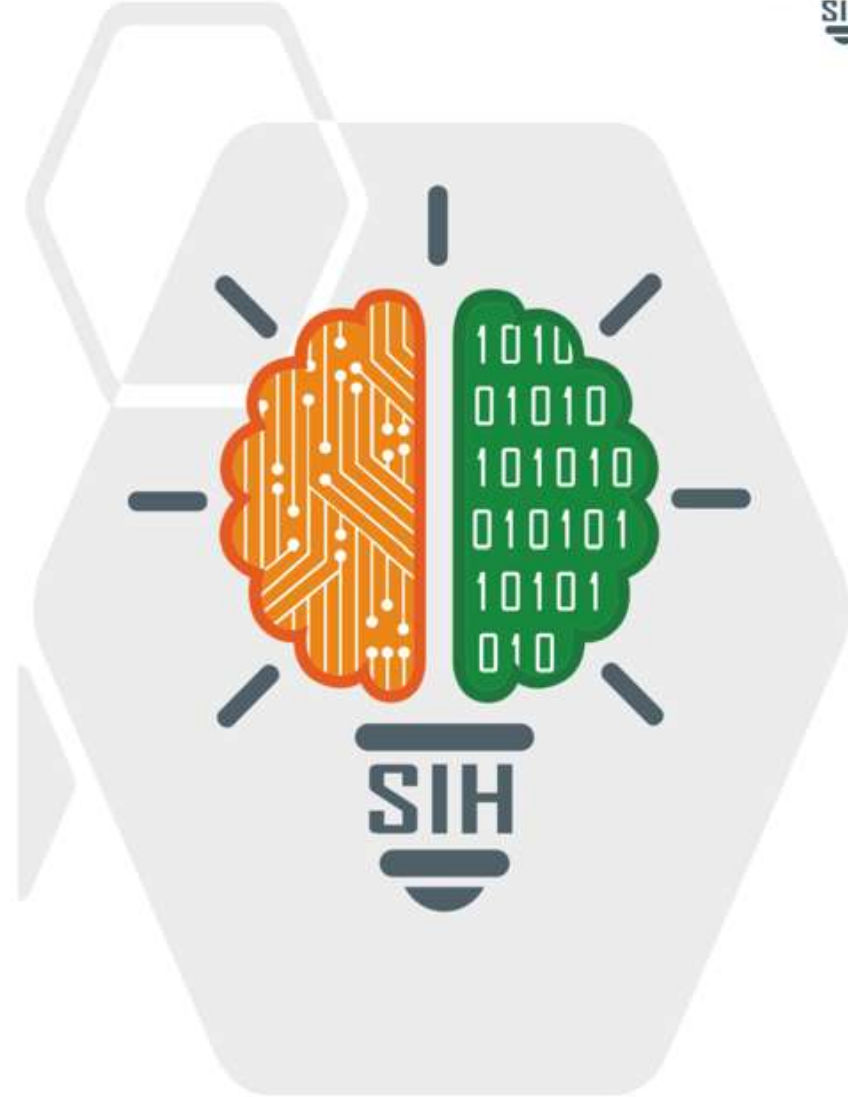


SMART INDIA HACKATHON 2026



- Problem Statement ID – 26219
- Problem Statement Title –
Soft Robotic Smart Feeding Assist Arm
for the Upper Limb Impaired
- Theme – Smart Automation
- PS Category – Hardware
- Team ID – 161201
- Team Name – LD Pvt Ltd (Limitations Denied Pvt Ltd)





Soft Robotic Smart Feeding Assist Arm for the Upper Limb Impaired

Assistive Robotics

Adaptive Grasping

Vision-Assisted

Independent Living



SMART INDIA
HACKATHON
2026

The Problem

- Caregiver assistance becomes necessary.
- Hand coordination remains impaired.
- Universal vessel adaptability needed.
- Social acceptability matters.



Target Users



Innovation: Material, working and usage.

Materials



Working Principle

- Pneumatic grip conforms to utensil shape
- No utensil drag
- Adapts to various shapes and sizes
- Maintains stable grip without dragging

Usage

- Works with plates, bowls, containers and different shapes
- Used in daily feeding tasks
- Integrated with vision and control system for precise and safe feeding

Idea

Self-Feeding Assistive Robot one arm · two control modes · soft grip

A Foot joystick + vision



- Feet steer the arm
- Base, elbow, spoon tilt (3-4 DOF)
- Camera tracks the mouth
- Stops if face is lost

Users with working legs

B Vision only



- Camera finds the mouth
- 3D mouth position (landmark detection)
- Mouth open 3 s = bite
- Stops if face is lost

Users without leg control

★ Soft robot ring

- Air-filled silicone ring (soft, flexible grip)
- Grips own bowl, plate, cup
- No special tableware
- Dish stays in place



Novelty - both control modes

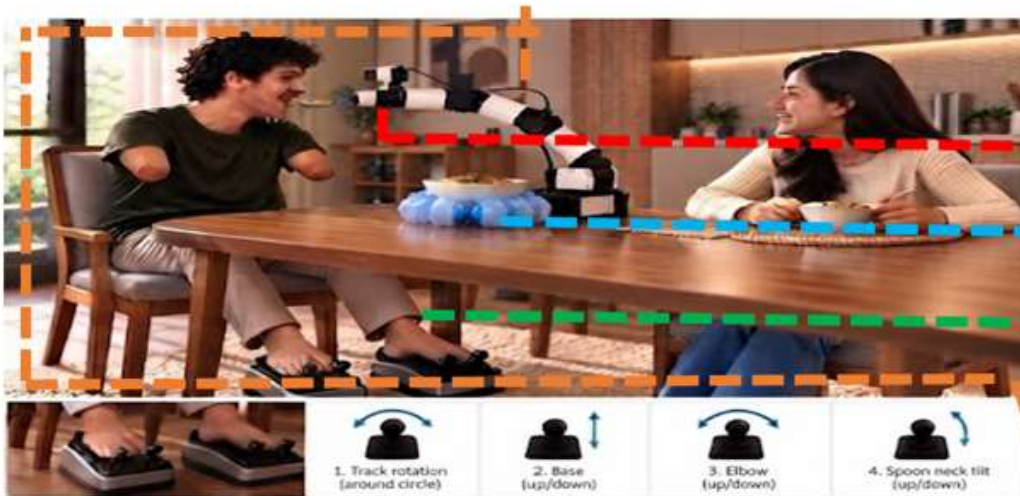
Regulations

CDSO · MDR 2017

ISO 13485

IEC 60601

FDA · EU MDR



<https://github.com/albertblessanth65/safe-feeding-assist-robot.git>



<https://youtu.be/QT2HadeyrNM>



Endorsement of: AICTE and Ministry of Social Justice and Empowerment, Government of India



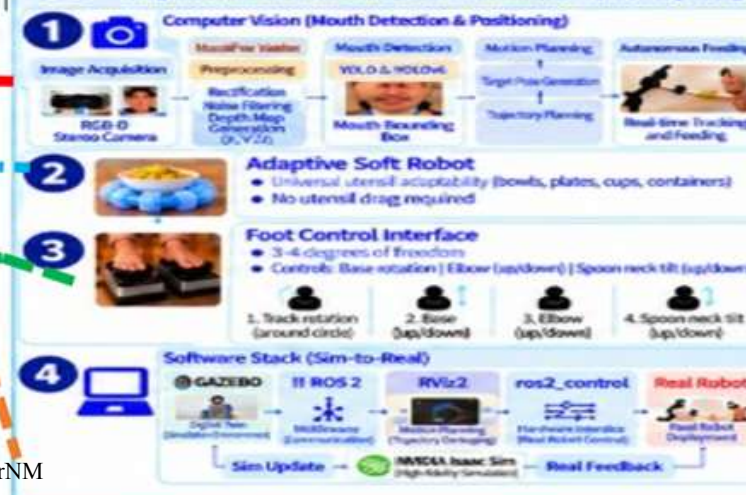
Department of Empowerment of Persons with Disabilities (DEPwD)



All India Council for Technical Education

Proposed Solution

Integrated system with user control, vision, robotics and adaptive soft grasping.



GITHUB LINK

<https://github.com/albertblessanth65/safe-feeding-assist-robot.git>

Website Link :

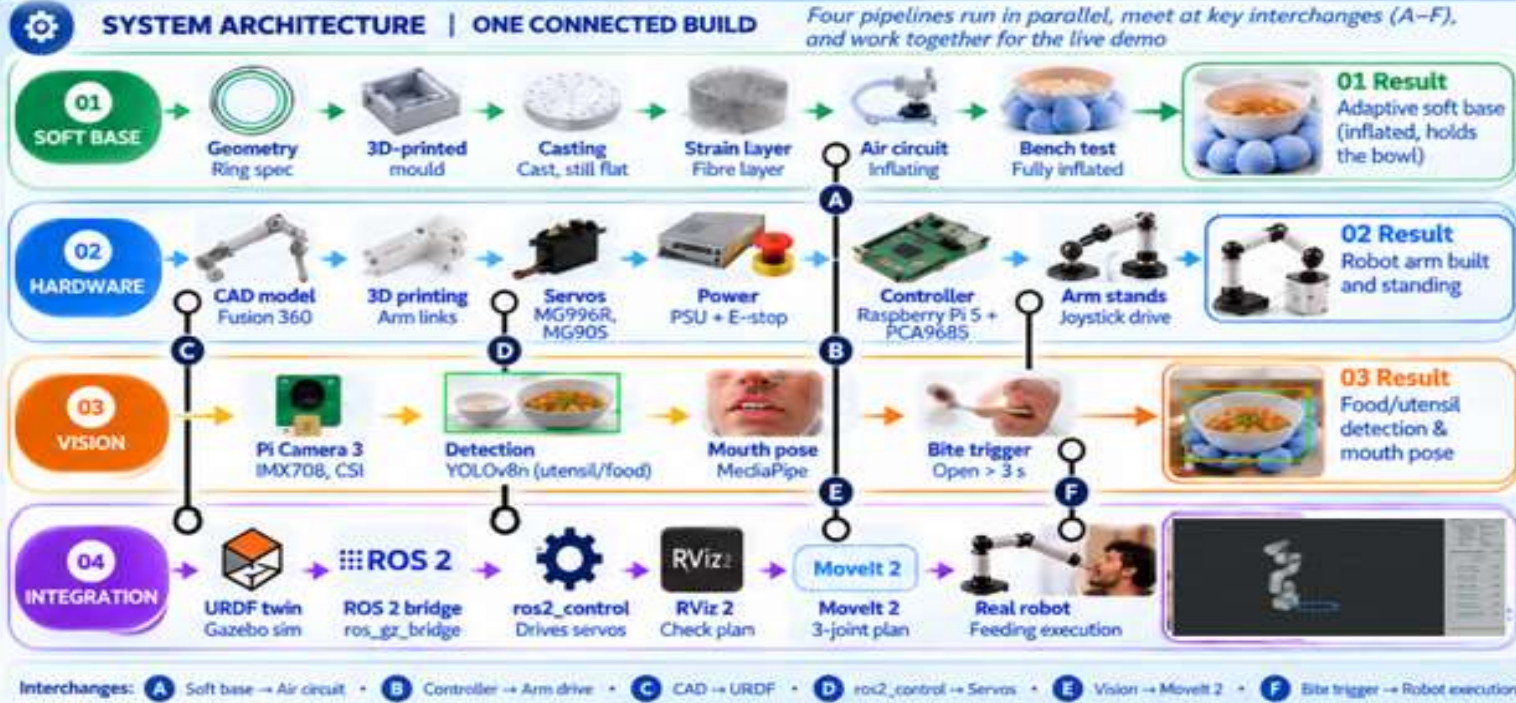
<https://caliburrito72.github.io/LD-Pvt-Ltd/>

Youtube Channel :

<https://youtu.be/QT2HadeyrNM>



TECHNICAL APPROACH



SYSTEM OUTPUT



Adaptive Soft Base
Inflated and stable



Vision Output
Food & utensil detection



Robotic Arm Execution
Feeding the user



PROJECT LINKS



GitHub Repository

<https://github.com/albertblessanth65/safe-feeding-assist-robot.git>



Demo Video

<https://youtu.be/QT2HadeyrNM>



GITHUB LINK

<https://github.com/albertblessanth65/safe-feeding-assist-robot.git>

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FEASIBILITY & VIABILITY



CHALLENGES & RISKS

01

Soft-base fatigue

Seam leak 90-95% accuracy [1][2]

02

Pi 5 vision

~15 FPS No depth [3]

03

Face safety

No feedback Hobby servos

04

Regulatory gate

CDSO class? Uncertified silicone [5]

05

sEMG misfires

Random twitch Unwanted motion

Threshold + hold

Mean + 3 SD Hold >150 ms [4]



HOW WE OVERCOME

01

One-piece casting

Cavity leak test Swap fingers

02

Fixed pose

Camera-set height D435 later

03

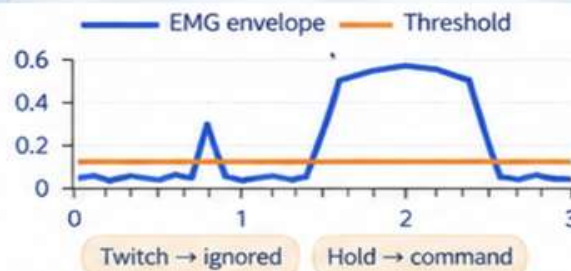
Layered safety

6 V rail E-stop Speed limit

04

Early compliance

CDSO enquiry Natural silicone



BUSINESS MODEL CANVAS



KEY PARTNERS

- Rehab centres
- OT therapists
- DePwD
- AIIMCO*
- Parts suppliers
- Engg. colleges



KEY ACTIVITIES

- Prototype build
- Soft casting
- OT trials
- CDSO filing



KEY RESOURCES

- Arm CAD
- ROS 2 stack
- Vision model
- Core team



VALUE PROPOSITION

- Self-feeding
- Own utensils
- Soft grip
- Joystick / vision
- sEMG ready
- ₹30.7k parts
- vs US\$6.5k+ [6]



CUSTOMER RELATIONSHIP

- Demo → setup
- Carer training
- Remote support



CHANNELS

- Rehab centres
- Disability NGOs
- Carer referrals
- ADIP camps*

1

CUSTOMER SEGMENT

- Stroke survivors
- Upper-limb amputees
- Carers
- Rehab centres
- ADIP users*



COST STRUCTURE

- Parts ≈ ₹17k
- Soft casting
- Engineering
- Testing
- CDSO compliance

5

REVENUE STREAM (Buyback Model)





IMPACT AND BENEFITS



POTENTIAL IMPACT



Social
Inclusivity



Independent
Living



Caregiver
Relief



Quality
Life



Key Ministries and Stakeholders



DEPWD
(Divyangjan)



Ministry of
Health and
Family Welfare



Ministry of
Science and
Technology



Ministry of
Education



Ministry of
MSME



NITI Aayog



Government Support



Funding



Policy



R&D Grants



Public Programs



BENEFITS

1 Competitors

Competitor Analysis



Obi (Gen 3)



I-Eat Feeding Robot



IIT FoodBuddy

1. Competitors and Cost

Product	Origin	Status	Cost
Obi (Gen 3)	USA	Commercial	~ ₹10-12 lakh per set - (USD 13126, Résem et al. 2023)
I-Eat Feeding Robot	USA	Commercial	~ ₹ 6 %
IIT Gauthami Netravayi	India	Patented, but not a startup yet	₹ 3,000 - ₹ 4,000

3 Market Analysis

India Market

9.2%
CAGR
(2024 - 2030)



- Large population
- High unmet needs
- Strong government push
- Growing market

Future Scope

- EMG / sEMG
Detect muscle signals to control leg
- CYBER SECURITY
Secure cure data and access
- SG INTEGRATION
Mingle Ingenl based based
- NG INTEGRATION
Innex based operation

2 How We Differ

Feature	Obi	I-Eat	FoodBuddy	Ours
Uses the user's own utensils	×	×	×	✓
Soft grip conforms to utensil shape	×	×	×	✓
Holds plates, bowls, cups, containers	×	×	×	✓
Locked to a proprietary tray or scoop	Yes (multi-tary)	Yes (single scoop)	Yes (single scoop)	Yes
Computer-vision robot	×	×	×	✓
Foot control	Yes (novel)	Yes (foot switch)	No (voice)	No (planned)
EMG/EEG controlling	×	×	×	Planned



Why India Can Lead



- Large population
- High unmet needs
- Strong government push
- Growing market



If we don't do it,
someone else will.



RESEARCH AND REFERENCES



SMART INDIA
HACKATHON
2026

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User Control

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02 SOFT ROBOT

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Shape Adaptation

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Soft Actuation

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Pneumatic Motion

Shintake et al. • 2018

Soft Grippers

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Flexible Grasping

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Leg-Based Control

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Myoelectric Input

Lyons & Joshi • 2018

Leg Gestures

<https://doi.org/10.1109/TNSRE.2018.2807360>

Prosthesis Control

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EMG + Flex

<https://doi.org/10.1109/ICEEET64463.2024.10866485>

Assistive Control

LD Pvt Ltd -Website



With our CoPilot — He kept the Show happening and made us laugh, and also gave Review of our work



<https://github.com/albertblessant/h65/safe-feeding-assist-robot.git>

Website Link :

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